



Bioinformatics for engineers

# Genome Assembly

Msc. Juan Sebastian Malagón Torres

Teaching assistant. Eliana Zuluaga Aponte

**Universidad Javeriana**



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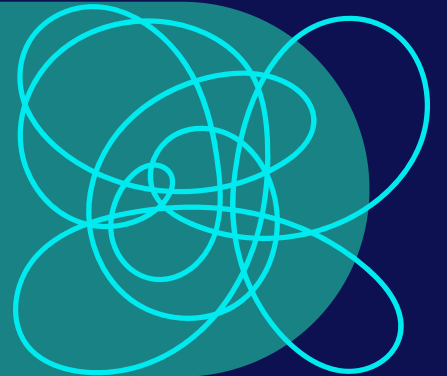
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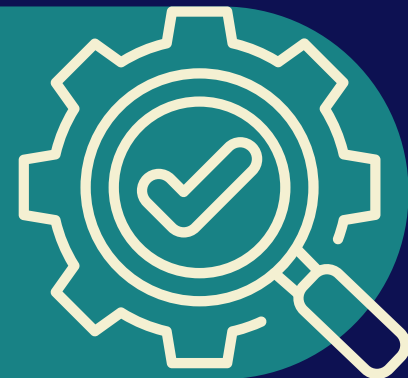
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Query ID

Nombre del gen

Preferred\_name

Nombre común de la proteína

Description

Descripción textual de la función de la proteína

GO Terms / KEGG  
Pathways

Códigos necesarios para hacer análisis de rutas  
metabólicas

# **WHAT IS GENOME ASSEMBLY?**

# Imagine you're extremely bored...

You are SO deeply and extremely bored that you take an entire biology book, rip all of its pages, cut each word from each page and THEN, the wind blows and you have to reorganize the entire book content!!!



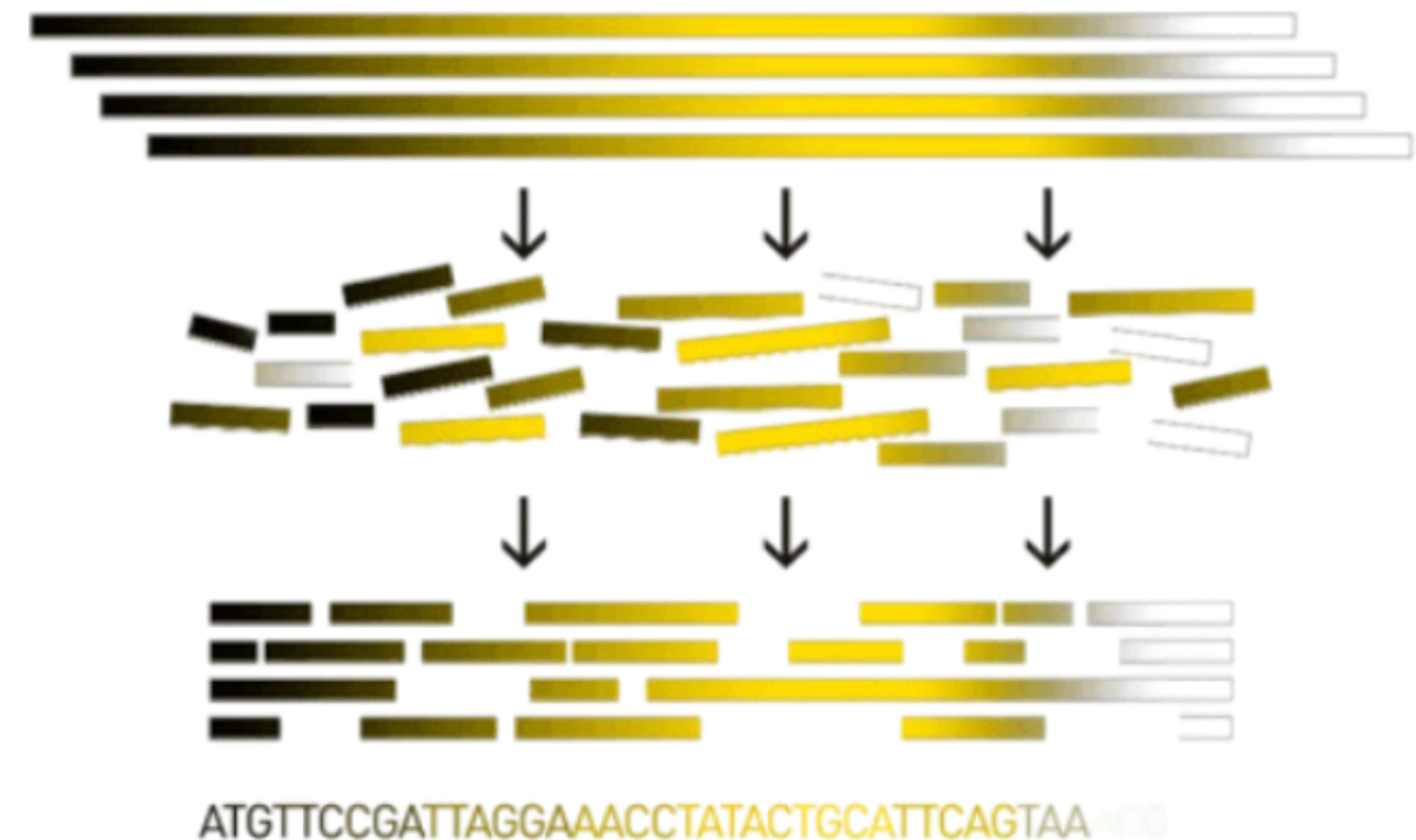
Exciting right?

# Imagine you're extremely bored...

Just kidding, but that is a good approximation of what we do in genome assembly.

Now your book is a complete genome. Your goal is to reconstruct as close as possible the complete genome, from sequences of DNA fragments (the "reads").

Genome assembly consists of aligning and reconstructing these fragments to form a continuous sequence (that of the chromosomes) or a set of contiguous sequences (called contigs or scaffolds).



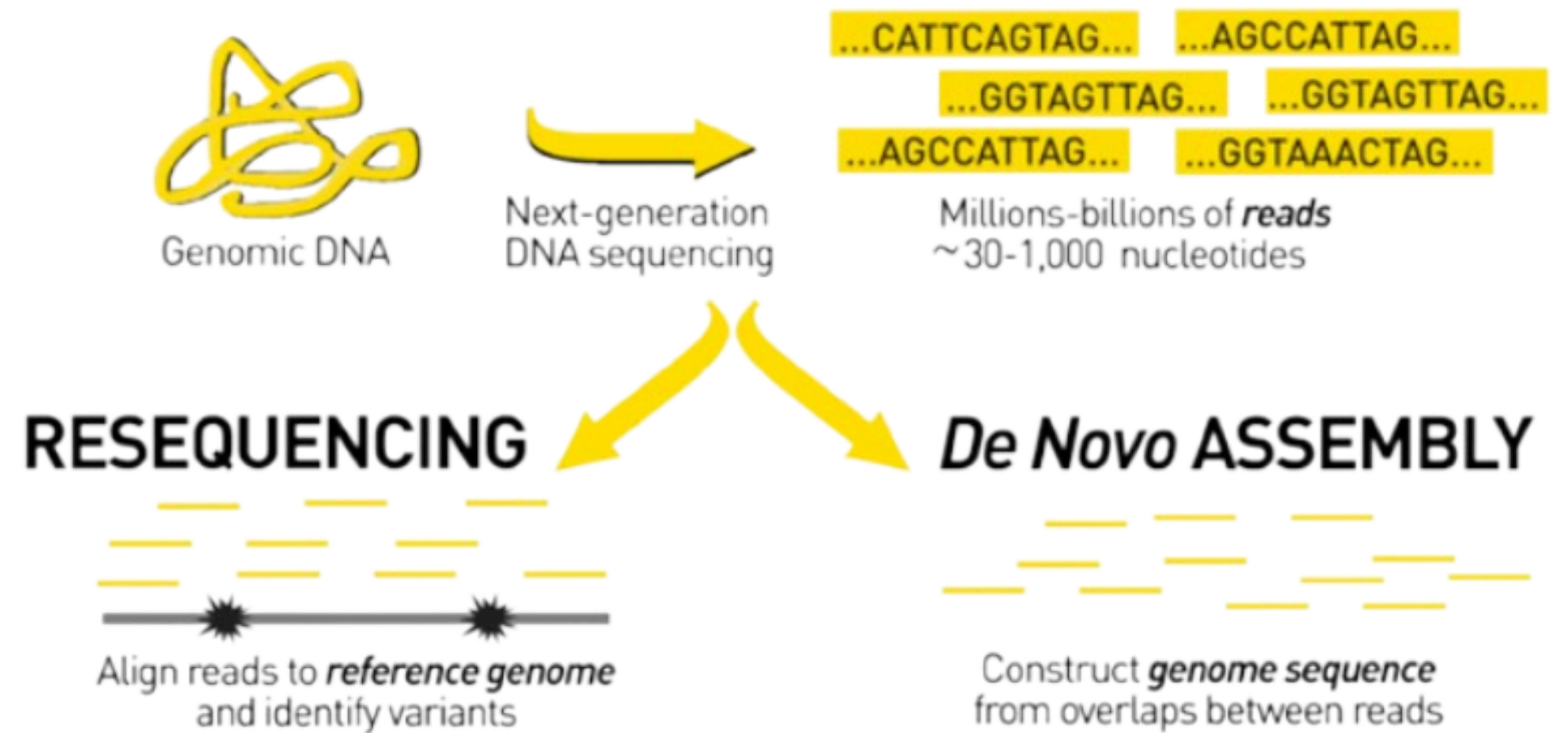
<https://www.hudsonalpha.org/sequencing-from-scratch-reference-genomes-and-de-novo-sequence-assembly/>

# **ASSEMBLY VS. ALIGNMENT**

# Genome assembly vs. alignment

When we worked with alignment methods we started with a known reference genome. The reads obtained by sequencing are "mapped" against this reference to determine their origin and identify variations, gene expression, etc.

But if we want to build a genome we depend on the existence of a good quality reference, this introduces bias towards the reference genome.



<https://www.hudsonalpha.org/sequencing-from-scratch-reference-genomes-and-de-novo-sequence-assembly/>

# How can you reconstruct a genome?



Reference assembly maps reads to a reference genome by identifying reads with similar nucleotides to the reference. Essentially a jigsaw puzzle.



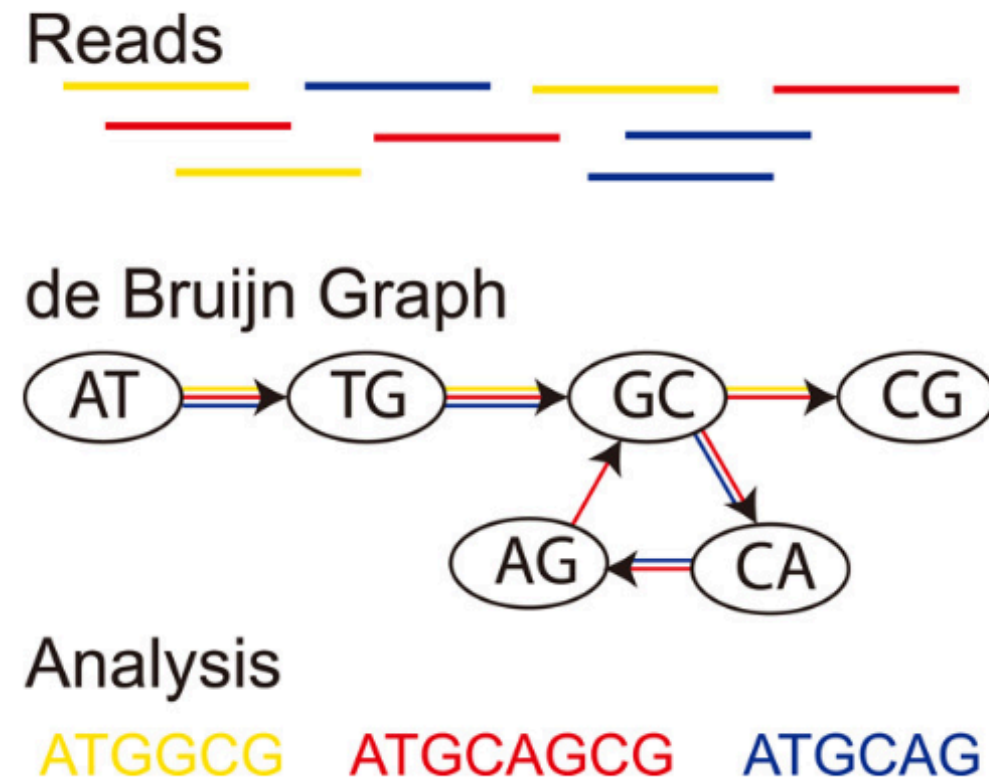
De novo assembly attempts to join reads together like a jigsaw puzzle but without a reference to compare reads to.

Today we'll focus on this one

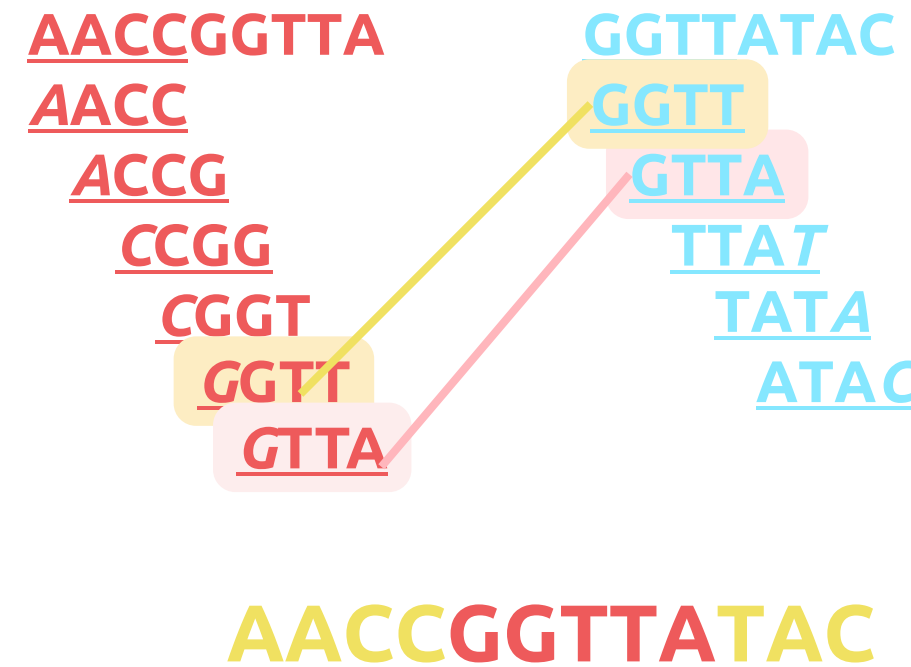
# DE BRUIJN GRAPHS

# De bruijn graphs

A De Bruijn graph is a directed graph, that represents **overlaps between sequences of symbols**. In bioinformatics, we can use this approach



We take k-mers out of reads to try and build an assembly. Like the example below:



These are all the possible 4-mers for these two particular sequences.

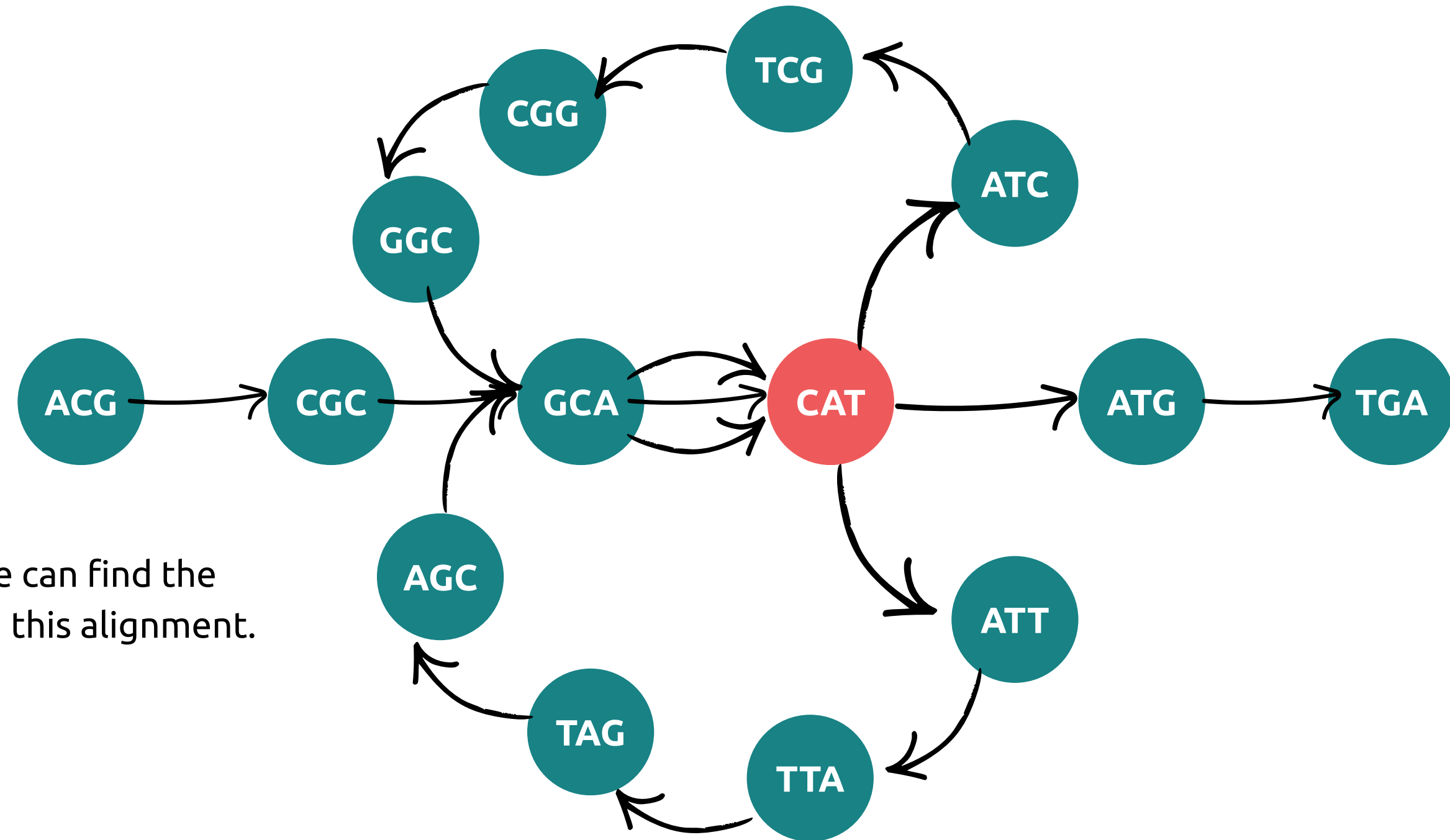
Now we have to identify the k-mers that overlap between the two sequences

As you can see, the overlap is clearly displayed when using the k-mers approach.

In reality, 4-mers occur very frequently, you would get them everywhere by chance, so you would have many coincidences, but a very low specificity (noise) and a poor assembly

This is why k must be selected wisely, one of the most popular De Bruijn assemblers, SPAdes, uses a combination of k-mers from 31 to 127

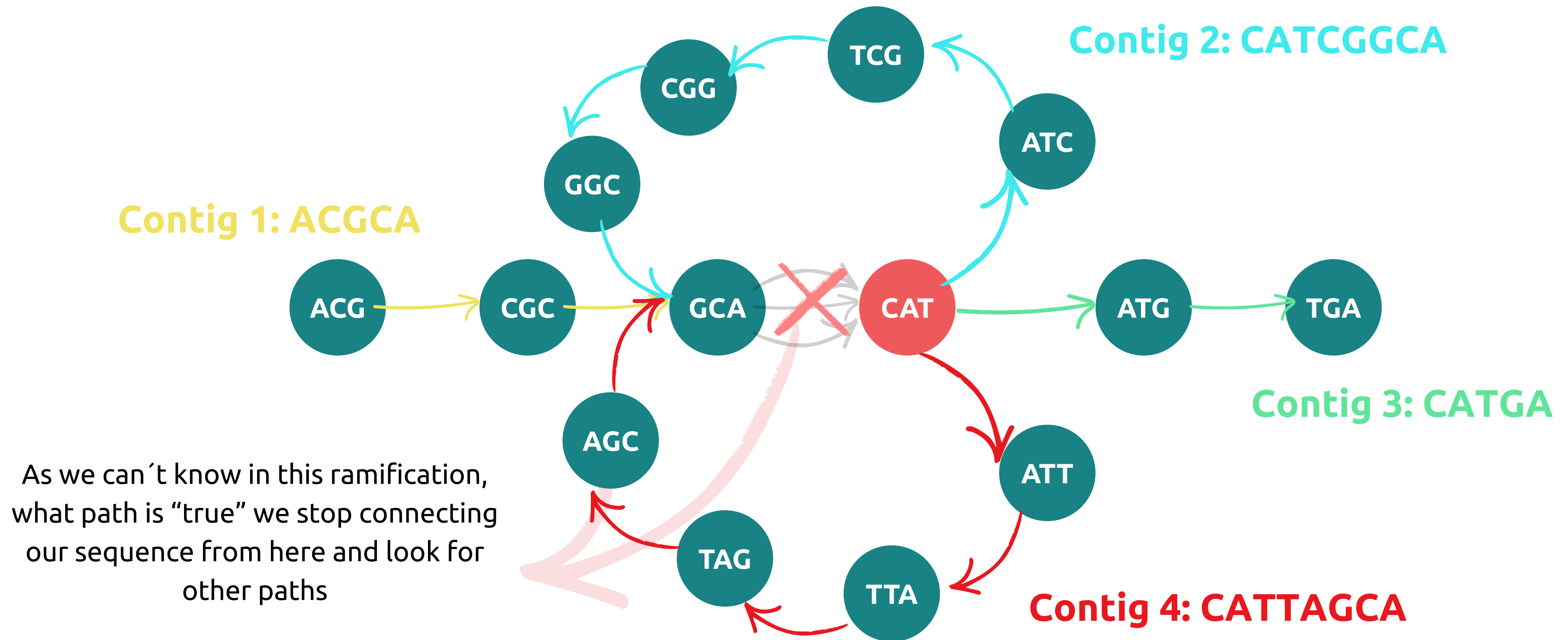
# De bruijn graphs



Out of this graph, we can find the contigs that form from this alignment.

# De bruijn graphs

Visualize what we mean by a directed graph



# De bruijn graphs

What problems we can get?

Going back to a simpler example, not nitrogenous bases.

**MISSISSIPI**

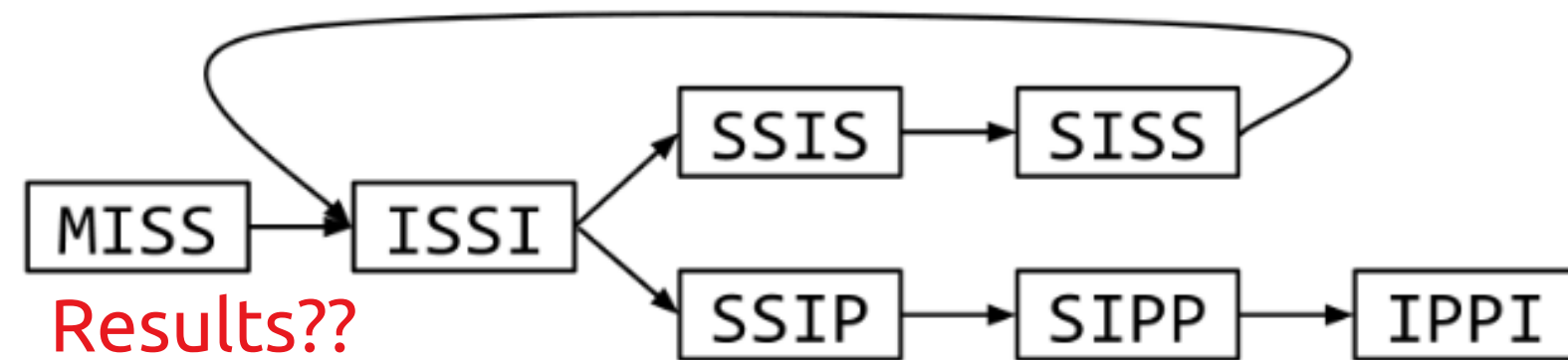
Using 4-mers we would get these:

MISS SSIS SSIP  
ISSI SISS SIPP  
SSIS ISSI IPPI

but... are they unique?

Umm... No... only seven are unique

So if you build your de Bruijn graph you wouldn't be using your entire word



Results??

MISSISSIPPI??? MISSISSISSISSIPPI??

Changing the number of K in your k-mers would give you many different results..

**CHOOSE K WISELY**

Lower K's

- More connections
- Lower chance of solving small repeats (Remember mississippi)
- Higher K-mer coverage

Higher K's

- Less connections
- Higher chance of solving small repeats
- Lower K-mer coverage

The length of your reads will also depend on your sequencing technology, this is something you should also consider

# QUALITY CONTROL



# How do we know if our assembly was good enough?

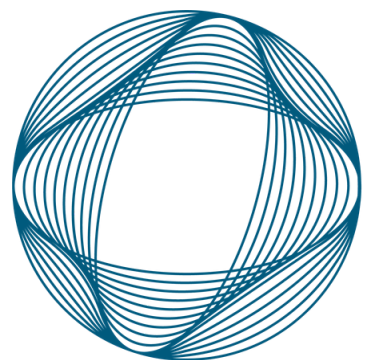
## 1. Continuity measures

How fragmented is the assembly?

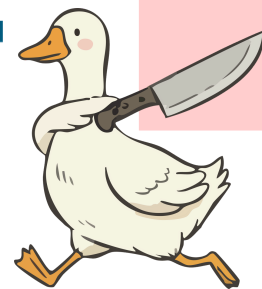
**N50:** This is the most used metric. If you order all the contigs from largest to smallest and sum their lengths, N50 is the length of the contig at which you reach 50% of the total assembly size. A high N50 indicates a more continuous assembly.

This is a summary of other continuity measures

| Metric               | Measures                                                        |
|----------------------|-----------------------------------------------------------------|
| N50 / N90            | Length of the contig that represents 50% or 90% of the assembly |
| L50/L90              | Minimum number of contigs that add up to 50% or 90%             |
| Longest contig       | The largest continuous fragment obtained                        |
| Total contig numbers | Fewer contigs is usually better                                 |



QuaST



# How do we know if our assembly was good enough?

## 2. Genome completeness

Is the entire genome represented?

# BUSCO

BUSCO (Benchmarking Universal Single-Copy Orthologs): searches for highly conserved single-copy genes in the organism's lineage. If most are complete, the assembly accurately captured the coding regions. It reports genes as: complete, duplicated, fragmented, or missing.

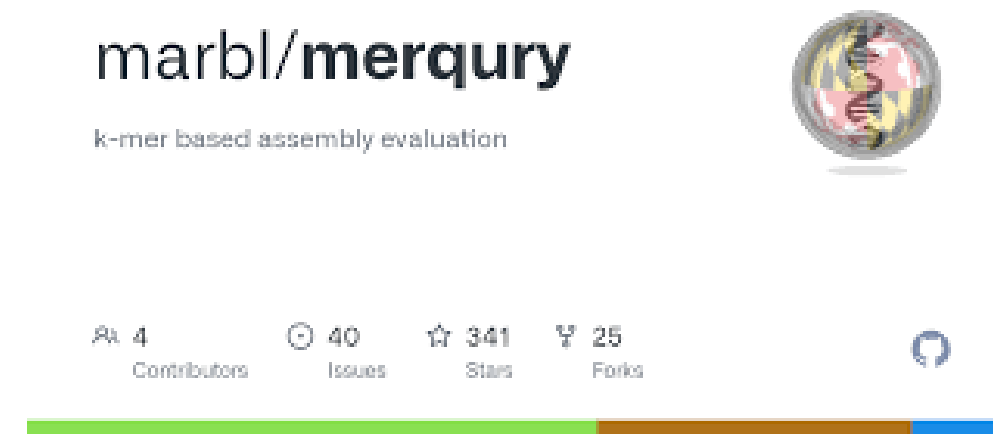
## 3. Sequence accuracy

Are the assembled nucleotides correct?

**QV (Quality Value):** estimates the error rate per base in the assembly.

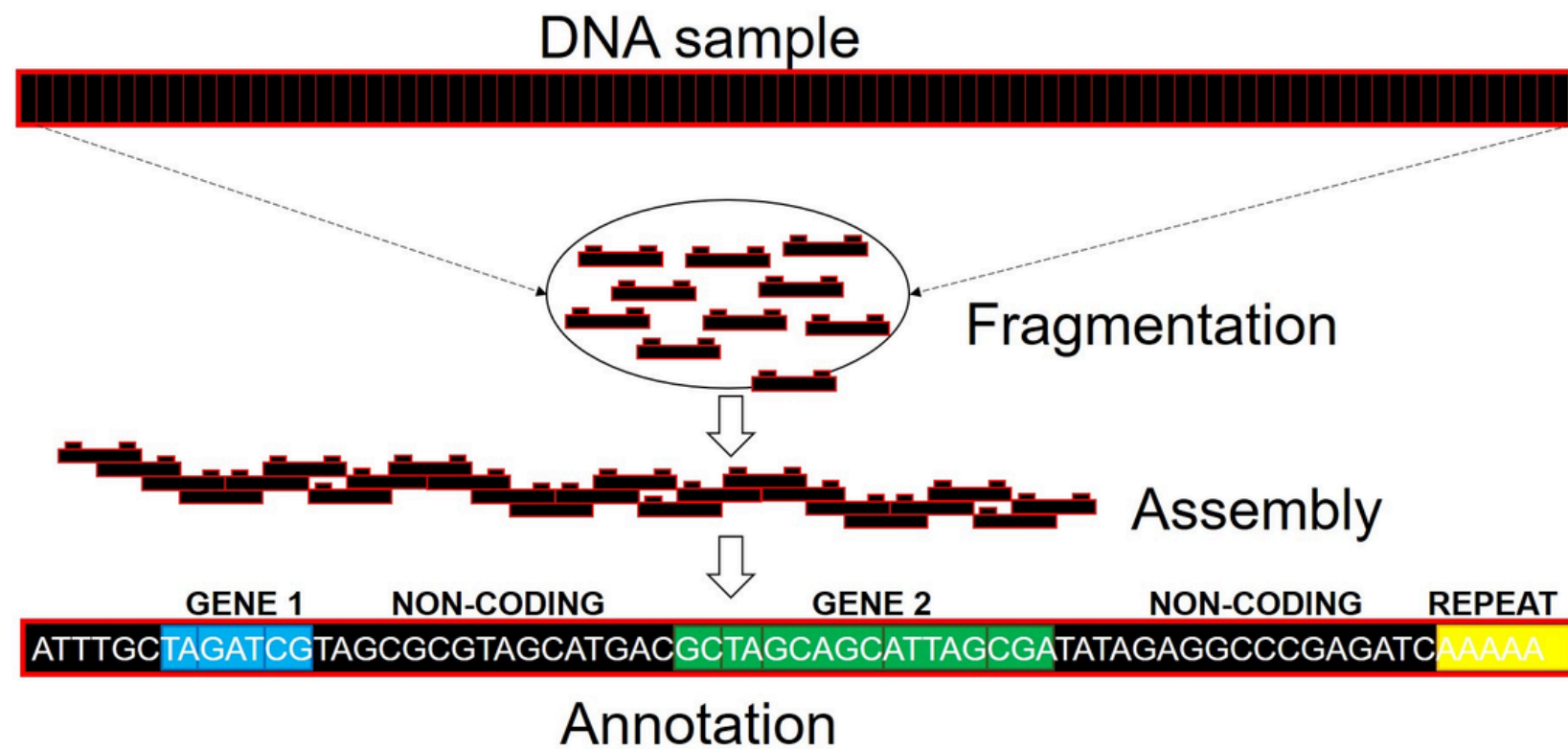
**k-mer analysis:** compares the k-mers of the assembly with the raw k-mers of the reads; detects erroneous bases, collapsed haplotypes, or duplicates.

It is calculated using tools like Mercury.

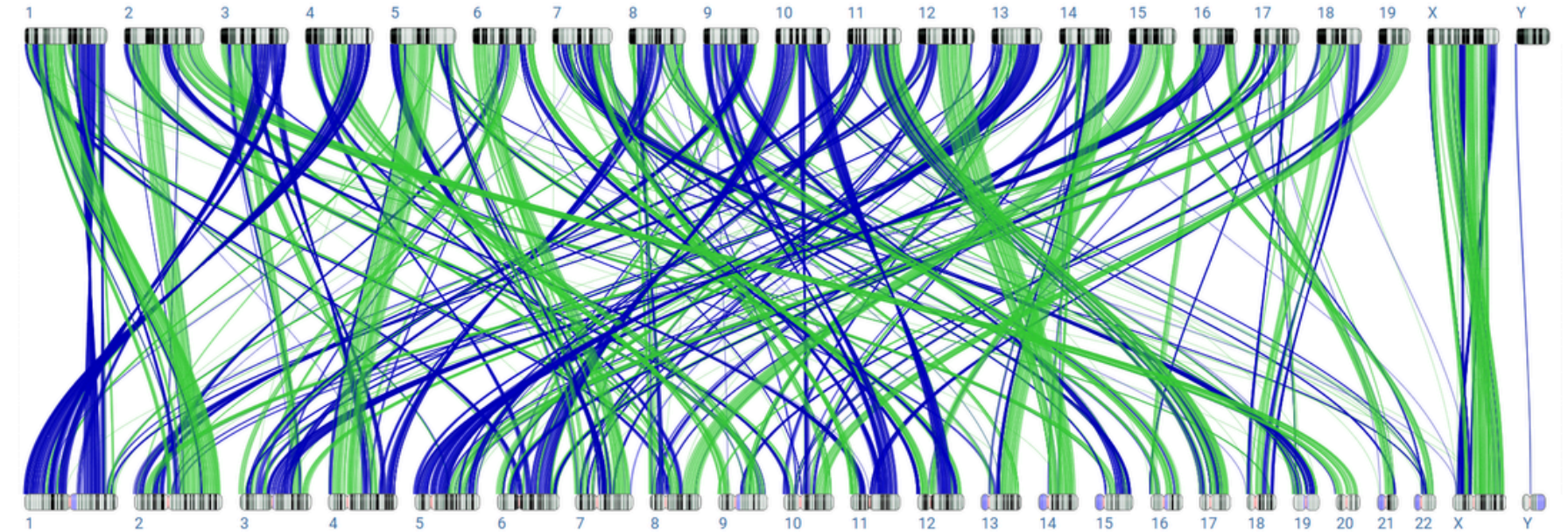


# NEXT STEPS

# A little spoiler



Remember what we did in GenBank?



## Genomic Annotation (From Sequence to Function)

The assembly process only gives us a giant string of letters (ATCG). The next necessary step is to use bioinformatics tools to "read" that text and find where the genes begin and end, and what proteins they produce.

## Comparative and Evolutionary Analysis

Once you have your genome assembled (for example, from a bacterial strain isolated in the laboratory), you can compare it with other databases to see mutations, antibiotic resistance, or how closely related it is to other species.

**Thanks !!!!!**